

Penny-Wise and Pound-Foolish in AI-Generated Image Detection

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Abstract—The rise of AI-generated images has sparked serious concerns about their potential misuse across various domains, prompting the urgent need for robust detection methods. Despite advancements, many current approaches prioritize short-term gains at the expense of long-term effectiveness. This paper critiques the overly specialized approach of fine-tuning pre-trained models for short-term gains on a single AI image dataset, while disregarding the long-term imperative of achieving generalization and knowledge retention. To address this trade-off issue, we propose a novel learning framework (PoundNet) for the generalization of AI-generated image detection on a pre-trained vision-language model. PoundNet incorporates a learnable prompt design and a balanced objective to preserve broad knowledge from upstream tasks (object classification) while enhancing generalization for downstream tasks (AI-generated image detection). We train PoundNet on a single standard AI image dataset, following common practice in the literature. We then evaluate its performance across 10 large-scale public AI-generated image detection datasets with 5 main evaluation metrics, forming the largest benchmark test set for assessing the generalization ability of AI-generated image detection models, to our knowledge. The comprehensive benchmark evaluation demonstrates that PoundNet successfully balances generalization with knowledge retention, achieving a remarkable relative improvement of 19% in AI-generated image detection performance compared to state-of-the-art methods, while maintaining a strong performance of 63% on object classification tasks.

Index Terms—AI-generated image detection, generalization, knowledge preservation, pre-trained vision-language model, learnable prompt design, balanced objective.

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I. INTRODUCTION

THE proliferation of AI image generation technologies (e.g., [1], [2], [3]) has raised significant concerns regarding its potential misuse in various domains, including politics, entertainment, and cybersecurity. AI-generated synthetic media, which either conveys entirely new content or convincingly depicts individuals saying or doing things they never did, poses a serious threat to the authenticity and trustworthiness of digital information and visuals. Consequently, there is an urgent need for robust AI-generated image detection methods to protect against potential malicious use cases.

In recent years, AI-generated image detection has made considerable progress, with numerous methods [4], [5], [6], [7], [8] being proposed to identify and mitigate their impact. Typically, an AI-generated image detector functions as a binary classifier trained on specialized deepfake datasets. Although a trained detector can perform well on known AI image generators (e.g., generative adversarial nets [9]), the challenge lies in generalizing to a wide range of unseen generation techniques [10], [11], [12]. To achieve generalization, as depicted in Fig. 1, previous works like [4], [13], [14] fine-tune pre-trained large-scale deep learning models, which excel at generalizing to new tasks, using a class-agnostic binary classification objective tailored to specific deepfake datasets. However, they frequently overlook the importance of balancing generalization ability with knowledge retention, which are two essentials for effective and efficient deep learning [15], [16]. This strategy is akin to being “Penny-Wise and Pound-Foolish”, where the penny represents the narrow objective of high accuracy on seen AI generators and the pound signifies the crucial long-term goal of generalization to unseen ones. Fine-tuning large pre-trained models [17] solely with a binary classification objective on downstream deepfake datasets is a “short-term gain” approach. While the trained detector can excel at identifying limited deepfake types encountered during training, it degrades the valuable, broad knowledge encoded in the pre-trained model. This aggressive specialization on the downstream task erodes the model’s upstream capabilities [16], [18], [19], thereby impairing its ability to generalize to unseen AI generators.

In light of the shortcomings associated with the greedy approach of fine-tuning pre-trained models on AI-generated image datasets, we propose a novel approach, PoundNet, that prioritizes both generalization and knowledge preservation. Illustrated in Fig. 1, our proposed approach differs from prior methods, which primarily focus on a class-agnostic binary

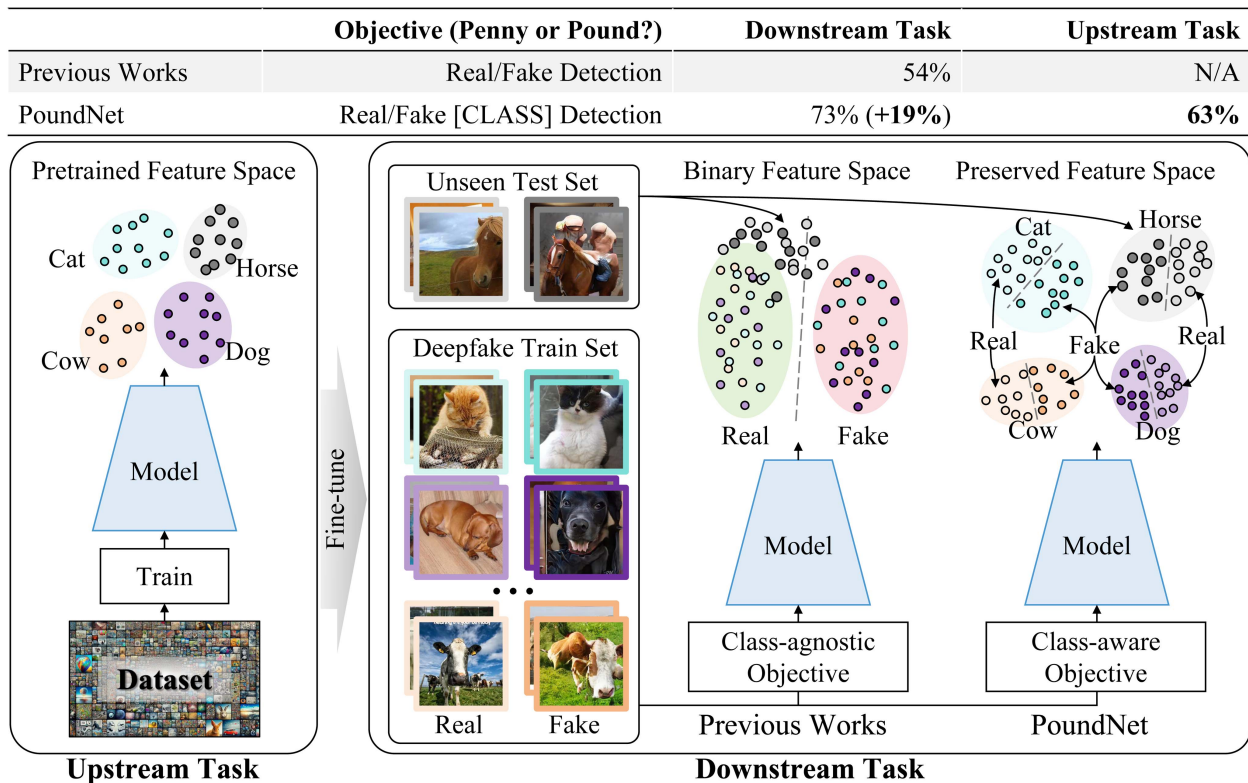


Fig. 1. “Penny” and “Pound” in previous methods and proposed approach (PoundNet) for AI-generated image detection and beyond. PoundNet improves AI-generated image detection (downstream) by 19% across 10 datasets and maintains a strong 63% performance in zero-shot object detection (upstream) across 5 datasets where many current detectors struggle.

training objective for AI image detection. Instead, PoundNet aims for class-aware AI-generated image detection, achieving notable improvements in generalization for downstream tasks (AI-generated image detection), with a substantial relative enhancement compared to state-of-the-art methods. Additionally, it demonstrates robust knowledge preservation, achieving a strong performance on upstream tasks (zero-shot object classification), a challenge for existing AI-generated image detection models due to their limited adaptability stemming from the greedy fine-tuning strategy. The efficacy of the class-aware deepfake detection objective in enhancing generalization is attributed to its simplification of the AI-generated image detection task by distinguishing between individual classes. Crucially, this class-aware objective is a training strategy that builds a more robust detector while ensuring it operates in a class-agnostic way, eliminating the need for re-training when encountering new semantic classes. Notably, large vision-language pre-trained models like CLIP [17], typically trained on extensive datasets (e.g., LAION-5B [20]), often encounter a broader range of classes during pre-training. Consequently, preserving class-based semantics during fine-tuning aids in generalizing to AI-generated image detection tasks. As shown in Fig. 1, the proposed PoundNet surpasses previous approaches by recognizing specific classes like ‘horse’. This focus reduces the primary task of distinguishing between real and fake instances within these classes, which is a more straightforward task compared to detecting fakes or reals without class information. Moreover, the class-aware objective improves the generalization of deepfake detection by preserving

the rich semantic knowledge from the pre-trained model during adaptation, which is the key to unlocking robust generalization for AI-generated image detection. Note that while we leverage upstream task performance (e.g., object classification) as a proxy to ensure knowledge retention, our primary goal remains the enhancement of downstream detection generalization.

In our experimental evaluation, we assess the performance of state-of-the-art AI-generated image detection models as well as our proposed PoundNet. The models are trained on a standard deepfake dataset, and their performance is evaluated on a wide range of unseen AI generators. Our results demonstrate the efficacy of our approach in achieving better generalization, showing around 19% improvement averaged across 5 standard metrics, compared to existing methods. Furthermore, our approach preserves prior knowledge for zero-shot classification on general datasets, achieving an accuracy of 63% on the ImageNet series benchmarks. In contrast, most state-of-the-art AI-generated image detection models are unable to perform upstream tasks. Overall, our proposed approach offers a more robust and reliable solution to the challenge of AI-generated image detection by balancing generalization capabilities and knowledge preservation. In summary, this paper’s main contributions include:

- Identifying the Penny-Wise and Pound-Foolish phenomenon within many current AI-generated image detection models.
- Proposing an Anti-Pennywise and Pound-Foolish learning framework (PoundNet) to improve generalization for detection (downstream), and maintain a promising

performance for classification (upstream) tasks in the context of fine-tuning pre-trained deep learning models.

- Evaluating the proposed PoundNet and 10 state-of-the-art AI-generated image detection methods using 10 large test datasets and 5 standard metrics, plus 5 datasets for object classification. To our knowledge, this benchmark is the largest for assessing deepfake detection generalization, as existing studies usually use one or two of these datasets.

II. RELATED WORK

A. AI-Generated Image Detection for Generalization

In recent years, many AI-generated image detectors [4], [21], [22], [23], [24], [25], [26] have been proposed for ensuring the authenticity of visual content. CNNDet [4] uses a plain binary cross-entropy loss to fine-tune a pre-trained Convolutional Neural Network (CNN) model, ResNet50 [27], for deepfake detection. The training dataset, ForenSynths, consists of real images from 20 categories in the LSUN dataset [28] with synthetic images generated by ProGAN [9]. They also propose using data augmentation, random JPEG compression, and Gaussian blurring to improve the model's cross-generator performance and robustness. Patchfor [29] utilizes truncation at different layers of neural networks (e.g., ResNet [27], Xception [30]) to obtain predictions based on the collective information from all patches. Recently, advancements in large pre-trained models, such as ViT [31] and CLIP [17], have demonstrated powerful zero-shot generalization across various computer vision tasks. UnivFD [13] utilizes the image features extracted by the pre-trained CLIP vision encoder to train a linear classifier for binary classification. Since CLIP already has extensive prior knowledge about the world, even training a detector exclusively on its frozen vision features can achieve good generalization. SPrompts [32] and CLIPping [33] employ prompt tuning strategies to fine-tune the CLIP model specifically for deepfake detection. When AI-generated image detection is treated as a downstream task for pre-trained models, previous methods that fine-tune these models with a binary classification objective on deepfake datasets adopt a "short-term gain" strategy. While effective for specific deepfake types encountered during training, this approach compromises the general knowledge in pre-trained models, hindering their ability to generalize to new AI generators and perform upstream tasks like object detection.

As synthetic images often lack high-frequency details compared to real images, [14], [34], [35], [36] learn to extract frequency information from images to achieve generalizability. LGrad [37] utilizes a pre-trained CNN model to convert images to gradients, which can extract generalizable features in GAN-generated images. Fusing [38] is a two-branch model that integrates global and local features of AI-synthesized images. Other methods [11], [39] use the predicted noise or reconstruction error of the image to learn consistent generalizable features. However, these methods are computationally expensive for inference, as they require a denoising model to preprocess test samples. All these methods can be regarded as using different feature extractors to obtain better generalizable features, but they still use the greedy binary classification objective for

training. Moreover, these methods heavily rely on low-level, imperceptible features within the image, which may not be robust in real-world scenarios. Recent works [4], [12], [40] have found that the effectiveness of low-level features can be easily compromised by common image processing techniques, such as JPEG compression.

Some recent studies [12], [41], [42] show that existing deepfake detectors trained solely on GAN-generated images do not effectively generalize to identify synthetic images produced by diffusion models. One reason is that the frequency spectrum of diffusion-generated images differs from that of GAN-generated images. This issue, along with the emergence of text-to-image generative models, underscores the need for more robust and adaptable detection methods. As the task becomes increasingly complex with the advent of more generative models, it is crucial to fully evaluate detection methods. To address this, several benchmarks [11], [12], [13], [43], [44] that contain mixed deepfake types for evaluation have been proposed. HiFi-Net [45] utilizes a multi-level classification system to enhance the detection and localization of various image forgeries, from Generative Adversarial Networks (GANs) to Diffusions. Some methods [19], [32], [46] have found that continual learning is an effective way to enhance the generalization ability for detecting unseen deepfakes by learning more deepfake types. However, none of the previous works explicitly frames this challenge as a trade-off between short-term gains and long-term costs. While they achieve "penny-wise" success on known artifacts, they often incur a "pound-foolish" penalty in generalization, preventing robust performance on new, unseen data.

B. Fine-Grained and Hierarchical Forgery Detection

To meet the demands of explainability and forensics, the research focus has shifted towards more fine-grained detection, which not only detects forgeries but also identifies their specific generation sources or properties.

Some fine-grained approaches formulate this as a multi-class attribution task. Yu et al. [47] find that distinct forensic traces are embedded in GAN-generated images and trained a multi-class classifier to attribute a forged image to its specific source GAN. This concept was further extended by De-Fake [48] to newer generative models, including diffusion models. Additionally, Guide-Space [49] constructs separate guide embeddings for real images and each specific forgery type (e.g., DeepFakes or Face2Face), enforcing a controlled separation between them via a specialized loss function. Despite their progress, these methods typically treat different forgery techniques as independent categories, overlooking the inherent relationships and connections that may exist between them.

To address this limitation, Guarnera et al. [50] proposed a three-level hierarchical classifier that first distinguishes real images from AI-generated images, then classifies the latter by model type (GANs and Diffusion Models), and finally identifies the specific generative architecture. Recently, HiFi-Net [45] and HiFi-Net++ [51] introduce a hierarchical fine-grained formulation where each manipulated image is described by a path in a predefined attribute hierarchy. HiFi-Net employs a multi-branch

feature extractor to capture this structure, while HiFi-Net++ further leverages the vision-language representations of pre-trained CLIP models with text prompts to enhance localization precision. This form of structured learning has proven to be a powerful prior, encouraging the model to learn representations that capture both broad forgery features and method-specific artifacts.

While these fine-grained and hierarchical methods primarily focus on forensic attribution, they typically require explicit labels of specific generator types. This reliance places them under a closed-set assumption, limiting their adaptability to unseen forgery types. In contrast, while our work shares the high-level motivation of moving beyond a simplistic binary paradigm to avoid overfitting, our fundamental goal is distinct. Rather than building a multi-class forensic tool, we aim to develop a highly generalizable binary detector robust to an open world of unknown forgeries. Instead of relying on forgery-specific labels, our approach leverages semantic categories to preserve pre-trained knowledge, thereby building robust generalization against unseen forgeries.

C. Prompt Tuning for Adapting Pre-Trained Models

Large vision-language models (VLMs), such as CLIP [17], have exhibited remarkable zero-shot performance across a wide range of computer vision tasks. Prompt tuning [52], [53], [54], [55] is a popular technique used to transfer pre-trained VLMs to specific downstream tasks without modifying the entire model. Instead of using hand-crafted prompts, CoOp [56] and Co-CoOp [57] optimize a set of continuous learnable vectors for downstream tasks, which are then concatenated with the input text prompts to adapt the VLMs to the specific task. VPT [58] extends the concept of prompt tuning to the visual domain, which uses learnable vectors as part of input images for vision encoder. MaPLe [59] enhances the performance and generalization capabilities of VLMs by leveraging multi-modal prompts of both textual and visual components. Recent works [60], [61] explore effective learning prompts from base datasets to capture rich, transferable knowledge across various domains. Similarly, SPrompts [32] finds that prompt tuning is an effective way to enhance generalization for detection. However, none of the existing studies explicitly investigates the trade-off between generalization and the preservation of pre-trained knowledge when adapting models for AI-generated image detection. In contrast, we propose a prompt-tuning strategy to train a robust AI-generated image detector by preserving learned knowledge to enhance generalization capabilities in deepfake detection.

III. PROPOSED APPROACH

A. Preliminary

Contrastive Language-Image Pre-Training (CLIP) [17] is a vision-language model that aligns images and text in a shared feature space through contrastive learning. The architecture includes a text encoder $\mathcal{G}$ and an image encoder $\mathcal{V}$, which map text and image features into a joint embedding space $\mathbb{R}^d$.

CLIP's generalization capabilities allow it to perform zero-shot classification. The class-specific text features are computed by the text encoder by accepting hand-crafted prompts with class names (e.g., "a photo of a [CLASS]"). The probability for the class c is given by

$$p(c|x) = \frac{\exp(\langle T_c, I \rangle)}{\sum_j^K \exp(\langle T_{c_j}, I \rangle)}, \quad (1)$$

where $I = \mathcal{V}(x)$ is the feature of the input image x , $T_i = \mathcal{G}(t_i)$ is the class-specific features of the textual input t_i for the class $c_i \in \mathcal{C}$, K is the class number, and $\langle \cdot, \cdot \rangle$ denotes the cosine similarity.

While CLIP performs well on general tasks, its ability to handle specific downstream tasks, such as AI image detection, is limited due to the abstract nature of concepts like "real" and "fake" being outside its training dataset. Employing CLIP for such tasks may require adaptation strategies, such as full fine-tuning or prompt tuning.

The proposed approach, PoundNet, aims to achieve class-aware AI-generated image detection while balancing generalization and knowledge preservation, leveraging CLIP's strong transferability to better handle unseen AI generators.

PoundNet builds upon CLIP [17] and fine-tunes it using our designed prompt pair and suggested balanced objective. Ideally, we can use the prompt format of "a [real/fake] photo of a [CLASS]" to guide the CLIP model in performing downstream binary AI image detection. However, it is challenging for pre-trained CLIP to understand the abstract concept of deepfake in natural language. As shown in Fig. 2 (Top Left), to better parameterize the "a [real/fake] photo" context for the AI-generated image detection scenario, we introduced learnable paired prompts for reals and fakes respectively. The learnable prompts, together with the class names, are input into the text encoder, which generates the corresponding text feature representations. On the other end, a training set of real and fake images is processed through the image encoder to generate associated image features. To align the text features and image features for AI-generated image detection and form a preserved feature space (Fig. 2(d)), we propose a balanced objective comprising three components: Class-Agnostic Binary term (Fig. 2(a)), which is a high-level, abstract concept, focuses on distinguishing real and fake instances without considering the specific semantic context. It represents a general binary classification of real versus fake. Semantic-Preserving term (Fig. 2(b)) ensures the retention of broader knowledge encoded within the pre-trained model, which is often neglected by existing approaches that excessively fine-tune using only class-agnostic binary term. Class-Aware Binary term (Fig. 2(c)) aims at discerning between individual classes (e.g., detecting fake images of cats within a set of cat images), making the AI-generated image detection process more streamlined and effective.

B. Prompt Design

A critical challenge in applying CLIP to AI-generated image detection is adapting its pre-trained visual-semantic knowledge, originally optimized for natural images, to effectively identify

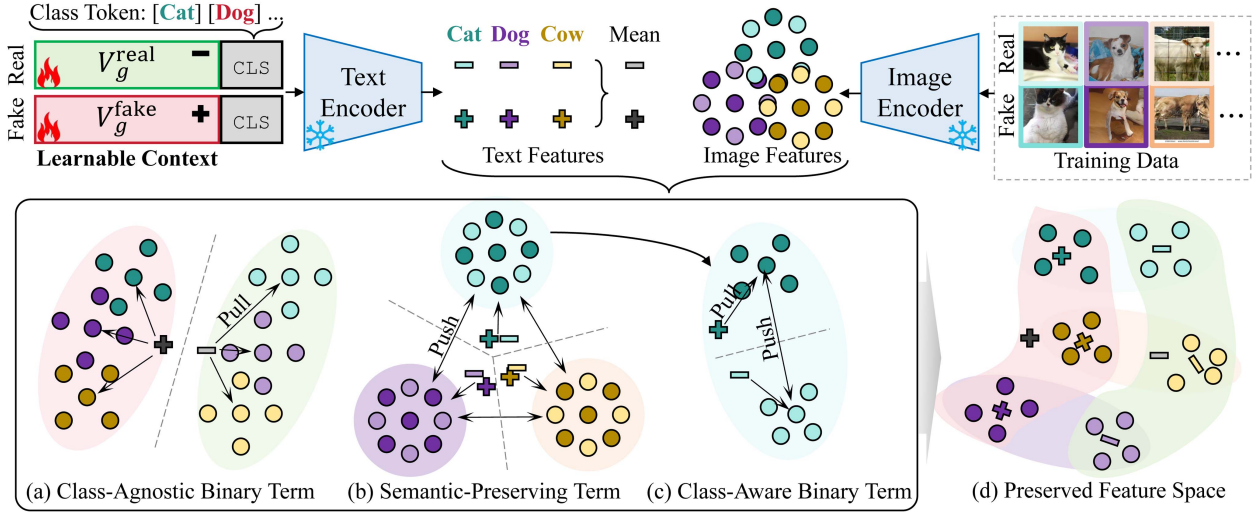


Fig. 2. Proposed framework (PoundNet) on a pre-trained vision-language model with a learnable prompt pair (Top Left) and a balanced objective (Bottom Left) with three basic loss components: (a) class-agnostic binary, (b) semantic-preserving, and (c) class-aware binary terms.

synthetic content. *How can we leverage CLIP’s rich multimodal knowledge and zero-shot capabilities to enhance the generalization for deepfake detection?*

Recent methods [32], [58], [59] use prompt tuning as an effective strategy for fine-tuning CLIP on downstream tasks to better preserve prior knowledge. For the language side, they suggest optimizing a set of M learnable continuous context vectors $V_G = [v_G^1, v_G^2, \dots, v_G^M]$ for downstream tasks. The input prompts for the text encoder follow the form $t_c = [V_G; c]$, where c represents the class token embedding corresponding to the class c . For deepfake detection, SPrompt [32] proposes appending the class names “real” and “fake” to a set of learnable prompts V_G and then doing optimization with a binary cross-entropy loss. Their prompts resemble hand-crafted versions like “a photo of [real/fake]”. However, similar to other prompting approaches, they cannot fully leverage the pre-trained knowledge of CLIP, since the concepts of “real” and “fake” are abstract and distant from the semantic concepts of general objects.

Instead of directly using “real” and “fake” as class names, we maintain the term [CLASS] in the prompts to represent general semantic classes and introduce a pair of learnable contexts, V_G^{fake} and V_G^{real} , to capture the abstract real and fake concepts, respectively. This design is analogous to using the phrases “a [fake/real] photo of a [CLASS].” in natural language, but instead of using hand-crafted textual descriptions, we optimize these context vectors through gradient backpropagation. As illustrated in Fig. 2 (Top Left), by concatenating the class tokens c with the context vectors (V_G^{fake} and V_G^{real}), and feeding these concatenated prompts into the text encoder, we can generate class-specific text features (T_c^{fake} and T_c^{real}) for both the fake and real versions of a given class c .

As discovered by [32], [59], it’s crucial to prompt both the vision and language branches simultaneously to enhance the efficacy of adaptation to downstream tasks. Thus, we follow them to additionally introduce a set of M learnable tokens as visual prompts $V_V = [v_V^1, v_V^2, \dots, v_V^M]$ into the vision branch, similar

to language prompts but combined with the image tokens. Both learnable text and vision prompts can be inserted into each transformer block of the encoder up to a depth J , allowing for more tuning parameters. During fine-tuning, we merely optimize prompts while keeping other network components frozen, which reduces the risk of overfitting and allows the model to adapt efficiently to new tasks without extensive retraining.

C. Balanced Objective

The training dataset for deepfake detection can be denoted as $\mathcal{D} = \{(x_i, y_i, c_i)\}_{i=1}^N$, where $x_i \in \mathcal{X}$ represents an image, $y_i \in \{0, 1\}$ is the binary label (0 for real, 1 for fake), and $c_i \in \mathcal{C}$ denotes a semantic category label (e.g., “dog”, “cat”) for image x_i . Semantic category labels are typically predefined in widely used AI-generated image datasets [4], [12] because they are created using generative models trained on datasets like ImageNet [62], LSUN [28], and CelebA [63].

The proposed balanced objective consists of three components: Class-Agnostic Binary term (Fig. 2(a)), Semantic-Preserving term (Fig. 2(b)), and Class-Aware Binary term (Fig. 2(c)). The Class-Agnostic Binary term is based on commonly-used training objective of previous work. It is the binary cross-entropy (BCE) loss function, as defined in (2). This loss function is class-agnostic as it only considers the binary labels y_i , disregarding the semantic category labels $\mathcal{C}$.

$$\mathcal{L}_{bce} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (2)$$

where $\hat{y}_i$ is the predicted probability that the image x_i is fake, and N is the number of samples.

To adapt $\mathcal{L}_{bce}$ to CLIP, which requires class tokens appended to the learnable contexts to make predictions as in (1), we propose an approach to achieve class-agnostic binary classification with semantic awareness. As shown in Fig. 2, we gather a set of class names $\mathcal{C}$ (e.g., “dog”, “cat”, “cow”, “etc.”) that adequately represent the diversity of objects or scenes the model

might encounter. Then, we concatenate the class tokens with the real and fake learnable contexts to obtain the corresponding text embeddings for the real and fake features of each class. We take the mean of all class embeddings to represent the general concept of real and fake, as $T^{\text{real}} = \frac{1}{C} \sum_j T_j^{\text{real}}$ and $T^{\text{fake}} = \frac{1}{C} \sum_j T_j^{\text{fake}}$. We can modify binary cross-entropy loss (2) by replacing $\hat{y}_i$ with $p(T^{\text{fake}}|x_i)$ as defined in (3).

$$p(T^{\text{fake}}|x_i) = \frac{\exp(\langle T^{\text{fake}}, I_i \rangle)}{\exp(\langle T^{\text{real}}, I_i \rangle) + \exp(\langle T^{\text{fake}}, I_i \rangle)}. \quad (3)$$

However, solely optimizing the binary cross-entropy loss $\mathcal{L}_{bce}$ for the downstream deepfake detection task can lead to a myopic focus on short-term performance gains. This approach fails to strike a balance between the downstream task and the valuable knowledge acquired from upstream pre-training tasks, resulting in a penny-wise and pound-foolish phenomenon. Thus, we further introduce a regularization loss called Semantic-Preserving Term $\mathcal{L}_{spm}$, which optimizes both context vectors, $\mathbf{V}_G^{\text{real}}$ and $\mathbf{V}_G^{\text{fake}}$, to perform general object classification, as shown in (4).

$$\mathcal{L}_{spm} = -\frac{1}{N} \sum_{i=1}^N \sum_{X \in \{\text{real}, \text{fake}\}} c_i \log \left(\frac{\exp(\langle T_{c_i}^X, I_i \rangle)}{\sum_j^K \exp(\langle T_{c_j}^X, I_i \rangle)} \right), \quad (4)$$

where $T_{c_i}^X$ is generated by the text encoder $\mathcal{G}$ with the input being $t_{c_i} = [\mathbf{V}_G^X; c_i]$, X is either for “real” or “fake”, and K is the total class number. $\mathcal{L}_{spm}$ only takes into account the semantic labels $\mathcal{C}$ and ignores the binary label y_i . The semantic-preserving objective acts as a crucial regularizer during fine-tuning. Conventional binary detection often leads to forgetting, where the model’s feature space collapses to focus solely on low-level, dataset-specific artifacts. By enforcing the retention of semantic knowledge (knowing “what” the object is), we compel the model to maintain a rich, structured feature space. This ensures that the detection of “real vs. fake” is grounded in a broader understanding of visual reality, rather than fragile, overfitting shortcuts. It ensures that the fine-tuning process maintains the semantic relationships and generalizable knowledge inherent in the original CLIP model. By doing so, we can leverage the rich contextual understanding of CLIP to enhance deepfake detection without compromising its ability to recognize and understand a wide range of objects and scenes.

In addition, we propose Class-Aware Binary Term, $\mathcal{L}_{cab}$, to smoothly transition CLIP from general object classification tasks to deepfake detection. $\mathcal{L}_{cab}$ is a binary cross-entropy loss applied within each category space, ensuring that the model can differentiate between real and fake images under the conditions of each class. $\mathcal{L}_{cab}$ can be formulated as

$$\begin{aligned} \mathcal{L}_{cab} = & -\frac{1}{N} \sum_i \sum_c^C (y_i \log p(c, y_i|x_i) \\ & + (1 - y_i) \log(1 - p(c, y_i|x_i))), \end{aligned} \quad (5)$$

where $p(c, y_i|x_i)$ is the probability that the sample x_i belongs to class c and has label y_i . $\mathcal{L}_{cab}$ ensures that the model learns to differentiate between real and fake images within the context of each semantic class, or in other words, in a class-wise manner. $\mathcal{L}_{cab}$ simplifies detection by shifting the learning objective from a

single abstract problem to a series of concrete, class-aware challenges. By learning to distinguish real from fake instances within the context of each object category, the model can effectively leverage its pre-trained semantic knowledge to identify subtle inconsistencies. This targeted, class-aware strategy is inherently more transferable and smoothly adapts CLIP’s feature space for the detection task with high efficiency.

During training, we optimize the PoundNet in an end-to-end manner with the balanced objective

$$\mathcal{L} = \mathcal{L}_{bce} + \lambda_1 \mathcal{L}_{spm} + \lambda_2 \mathcal{L}_{cab}, \quad (6)$$

where λ_1 and λ_2 are hyperparameters that balance the impact of each term. We simply set both λ_1 and λ_2 to 1 in our main experiments. Their impact is explored in the ablation study.

D. Inference

A key strength of PoundNet is its class-agnostic inference capability, which ensures broad generalization without the need for re-training on new semantic classes. For inference on images with unknown semantic classes, PoundNet generates paired text embeddings representing the general concepts of “real” and “fake”. We select a set of generic class names from datasets like LSUN or ImageNet. For each class c , the text encoder $\mathcal{G}$ processes two prompt variants: $t_c^{\text{real}} = [\mathbf{V}_G^{\text{real}}; c]$ and $t_c^{\text{fake}} = [\mathbf{V}_G^{\text{fake}}; c]$. This results in the embeddings T_c^{real} and T_c^{fake} . The mean of all class embeddings is computed to obtain T^{real} and T^{fake} . The final detection score is calculated based on (3). This approach ensures the detector remains effective without prior knowledge of the test samples’ groundtruth labels.

When the semantic class of input samples is known, leveraging class-specific embeddings ($T_{c_i}^X$) significantly enhances detection capabilities. This approach is particularly effective in scenarios with well-defined class labels, such as facial deepfake detection where the target class is explicit “face”. By incorporating these class-specific embeddings through the mechanism defined in (5), our model achieves superior detection accuracy. Our experimental results demonstrate that this method effectively captures and utilizes the subtle contextual differences between classes. Furthermore, PoundNet can also perform object classification by employing either $\mathbf{V}_G^{\text{fake}}$ or $\mathbf{V}_G^{\text{real}}$ as class-specific features, with predictions generated according to (1).

In summary, while PoundNet is trained in a class-aware manner to preserve knowledge and build a robust representation, it can function in the same way as traditional binary AI image detectors during inference.

IV. EXPERIMENTS

A. Benchmark Datasets

To thoroughly evaluate the generalization capabilities of our method and existing approaches, the evaluation is conducted on a large set of 10 public AI-generated image benchmarks, containing over 50 different AI generators.¹ For

¹ Although some generative models overlap, variations in creation lead to diverse AI-generated images, including different classes and styles, which are highlighted in the experiments.

general AI-generated image detection, we select 7 large public datasets for testing include ForenSynths [4], GenImage [12], GANGen-Detection [64], DiffusionForensics [11], Ojha [13], AntiFake [43] and DIF [44]. For classical deepfake detection, we select 3 large public datasets UADFV [65], Celeb-DF v1 and v2 [66]. To evaluate the zero-shot performance on general classification tasks, we employ the widely-used ImageNet dataset [62] and four of its variants: ImageNet-V2 [67], ImageNet-S [68], ImageNet-A [69], and ImageNet-R [70]. The detailed statistics of the benchmark results are provided in the Appendix, online available.

However, it's worth noting that PoundNet and other methods are only trained on the training set of ForenSynths [4], which is consistent with previous works [12], [13], [43], [44], [64]. We directly evaluate our models on the above-mentioned 10 benchmarks to simulate real-world scenarios, without any further fine-tuning. Although some generators, such as Stable Diffusion [1], are shared among the benchmarks, the varying difficulty levels of different test sets generated by the same generator can still be observed in the experimental results.

B. Evaluation Protocols

Data-side Bias: Recent methods [4], [14], [35] often assess their main performances without employing cropping, resizing, or compression techniques during the testing phase. However, as highlighted in recent studies [40], real images in most benchmarks [12], [13], [43], [44], [64] are predominantly stored in JPEG format, featuring a variety of sizes and aspect ratios, while their AI-generated counterparts are saved in the lossless PNG format with a uniform square aspect ratio. Given that JPEG is a common image format, these format differences should not be considered as inherent characteristics for distinguishing between real and AI-generated images. To mitigate this data-side bias, we convert all test images from 7 general AI-generated image detection benchmarks to JPEG format using a default quality factor of 90%. Since the three classical deepfake benchmarks already store data in JPEG format, we followed the standard process outlined by DeepfakeBench [71]. We make all our implementations open source to facilitate the replication of our evaluations.

Metric-side Bias: In the evaluation of deepfake detection, widely used metrics include the ROC Area Under Curve (ROC-AUC) and Average Precision (AP). These metrics can effectively reflect the sorting ability of a binary classifier and provide insight into its performance. However, we find that these metrics can't fully reflect the performance of detectors when applied to unseen generators, also known as out-of-domain data. Specifically, although some methods have similar areas under the precision-recall curve, indicating comparable AP or ROC-AUC scores, they may only maintain high precision and recall within a very narrow threshold range. The robust AI-generated image detector should perform consistently well across a broad spectrum of thresholds.

To address this issue, we design a new metric, AUC_{f1} , to reflect the threshold sensitivity of detectors in (8). This metric

calculates the area under the curve of the F1 score across different thresholds. By evaluating the F1 score over a range of thresholds, AUC_{f1} provides a more comprehensive measure of a detector's robustness and performance consistency. Similarly, we also use the Area Under the F2-threshold Curve (AUC_{f2}), which emphasizes recall over precision.

$$F1(\tau) = \frac{2 \cdot \text{Precision}(\tau) \cdot \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau)}, \quad (7)$$

$$AUC_{f1} = \int_0^1 F1(\tau) d\tau, \quad (8)$$

We also provide the widely used metrics including Average Precision (AP), Accuracy (ACC), F1 score (F1) in the experiments. All metric values are reported as percentages, with the '%' symbol omitted for brevity. For threshold-dependent metrics such as Accuracy and F1 Score, we report the results using a fixed decision threshold of 0.5 for all evaluated methods. This standard practice ensures a consistent and fair comparison across all models, as it avoids any model-specific threshold tuning that could potentially bias the results. We present the precision-recall curves with thresholds across all benchmarks and methods in the Appendix, online available. Note that AntiFake [43] uses separate splits for real images and deepfake types, making Accuracy the sole evaluation metric.

State-of-the-art Methods: We compare the proposed PoundNet against 10 state-of-the-art deepfake detection methods. These methods employ various techniques, such as analyzing frequency domain characteristics, learning inherent low-level patterns, and leveraging pre-trained models.

- CnNDet [4] employs a ResNet classifier trained solely on ProGAN-generated images to detect synthetic images across diverse generator architectures and models.
- FreDect [34] detects deepfake images by analyzing their frequency domain characteristics using Discrete Cosine Transform (DCT).
- GramNet [36] analyzes distinctive texture patterns in images and focuses on global statistical features to detect artificially generated faces with high accuracy.
- Fusing [38] combines global image features with carefully selected local patches through a two-branch neural network to effectively detect AI-generated images.
- LNP [39] focuses on learning the inherent patterns of real images to create a dense subspace, allowing for efficient identification of synthetic images that fall outside this space, regardless of their generative source.
- LGrad [37] detects GAN-generated images by analyzing gradient-based artifacts present in the images using a pre-trained CNN model.
- NPR [35] analyzes the unique local pixel interdependence patterns created during the up-sampling operations of CNN-based generators, enabling generalizable deepfake detection across various generation techniques.
- FreqNet [14] leverages frequency domain learning to analyze high-frequency information across spatial and channel

TABLE I

AVERAGE DEEPFAKE DETECTION PERFORMANCE OF THE PROPOSED POUNDNET AND 10 STATE-OF-THE-ART METHODS ON 10 PUBLIC DEEPFAKE DETECTION BENCHMARKS USING 5 MAIN EVALUATION METRICS (AP, F1, ACC, AUC_{f1} , AND AUC_{f2}). TABLE III PRESENTS THE RESULTS FOR THE FIRST FOUR EVALUATION METRICS ACROSS EACH DATASET, WHILE TABLES IN APPENDIX D, ONLINE AVAILABLE, REPORT MORE DETAILED PERFORMANCE WITH MORE METRICS AND ADDITIONAL CURVES. **BOLD** AND UNDERLINED HIGHLIGHT THE BEST AND THE SECOND-BEST PERFORMANCE, RESPECTIVELY. *NOTE THAT AP MERELY REFLECTS PERFORMANCE IN A MUCH SMALLER AND CLUSTERED RANGE COMPARED TO AUC_{f1} AND AUC_{f2} .*

	CNNDet	FreDect	GramNet	Fusing	LNP	SPrompts	UnivFD	LGrad	NPR	Freqnet	PoundNet
AP	72.53	69.23	60.69	75.05	62.42	76.36	<u>86.09</u>	56.82	54.41	56.22	86.14 (+0.05)
F1	17.10	<u>47.50</u>	9.87	16.87	10.23	16.26	<u>42.56</u>	1.38	14.63	13.36	70.58 (+23.08)
ACC	47.01	54.29	44.06	47.43	43.76	46.51	<u>58.90</u>	41.85	43.37	42.67	72.00 (+13.10)
AUC_{f1}	17.50	43.61	10.16	17.65	10.88	16.90	<u>43.10</u>	1.89	14.88	13.73	67.61 (+24.00)
AUC_{f2}	14.38	<u>42.28</u>	9.64	14.73	8.17	13.55	<u>37.69</u>	1.55	13.94	12.55	66.62 (+24.34)
Average	33.70	51.38	26.88	34.35	27.09	33.92	<u>53.67</u>	20.70	28.25	27.71	72.59 (+18.92)

TABLE II

ZERO-SHOT OBJECT CLASSIFICATION PERFORMANCE OF THE PROPOSED POUNDNET. CLIP USES THE OFFICIAL PROMPT TEMPLATES SPECIFICALLY DESIGNED FOR TESTING IMAGENET. PoundNet_{real}/PoundNet_{fake}: POUNDNET LEVERAGES CONTEXT VECTORS (TEXTUAL PROMPT) REPRESENTING REAL/FAKE CLASSES RESPECTIVELY, LEARNED FROM THE DEEPFAKE DATASET, TO PERFORM INFERENCE.

Method	ImageNet-A [69]	ImageNet-V2 [67]	ImageNet-R [70]	ImageNet-S [68]	ImageNet [62]	Average
CLIP	70.92	69.77	86.91	58.51	75.16	72.25
PoundNet _{fake}	57.67	57.92	79.63	48.32	63.96	61.50
PoundNet _{real}	60.40	60.21	79.62	49.11	66.01	63.07

dimensions, enabling superior generalizability across multiple GAN models.

- SPrompts [32] leverages prompt learning for both vision and language to fine-tune the pre-trained CLIP model, enabling generalizable deepfake detection.
- UnivFD [13] uses feature spaces from the pre-trained vision-language model, CLIP, and trains a simple linear classifier based on these features to achieve highly generalizable deepfake detection.

C. Implementation Details

Following [13], [32], we use the same pre-trained CLIP model [17], ViT-L/14, for all CLIP-based methods like PoundNet, SPrompts [32], and UnivFD [13]. The architecture of CLIP ViT-L/14 has 389 million parameters and has a storage size of around 1.4 GB in a PyTorch-compatible format. We insert a pair of learnable contexts, each with a length of 16, into the text encoder for both real and fake, respectively. To enhance tuning for downstream tasks, we also add prompts to the vision encoder, with a length of 16. The prompts we add to CLIP are negligible in size, consisting of only 241 K parameters.

PoundNet is trained for 1 epoch using the Adam optimizer [72], with a weight decay of 10^{-4} and a batch size of 128. We set the hyperparameters of λ_1 and λ_2 to 1 in the main experiments. We adopt the same data augmentations with previous works [4], [13] by applying Gaussian blurring and JPEG compression with a probability of 10% to the images during training. Using additional prompts does not increase the inference time for the CLIP model. PoundNet can process approximately 300 samples per second on an A100 GPU under full optimization. Following previous works [4], [32], we train PoundNet and all other methods on the training set of ForenSynths [4], where deepfakes are generated by ProGAN [9]. Training PoundNet takes about 1 h on an A100 GPU, with a single epoch.

D. Comparisons to the State-of-The-Art

We present the results of a comprehensive evaluation of the proposed PoundNet, comparing it to 10 state-of-the-art methods for deepfake detection. The overall evaluation results for deepfake detection and zero-shot object classification are presented in Tables I and II. The detailed evaluation results on 10 deepfake test datasets are presented in Table III. Except for the results related to the AP metric, which reflect performance within a much narrower and more clustered range compared to AUC_{f1} and AUC_{f2} as studied in Appendix, online available, Tables I and III show that PoundNet exhibits exceptional performance improvement over the other methods across the rest evaluation metrics and all the 10 used AI image detection datasets. Notably, our method achieves excellent results on traditional deepfake detection benchmarks (UADFV [65], Celeb-DF v1 and v2 [66]) without requiring any training on human faces deepfake datasets. This highlights its impressive generalization capabilities in the challenging domain of deepfake detection that requests strong generalization. Furthermore, Table II demonstrates that the proposed PoundNet still performs well for the zero-shot object classification tasks. The other evaluated detectors, due to their fine-tuning on a narrow, class-agnostic binary objective, are architecturally incapable of performing this semantic classification task. This sharp contrast indicates that existing binary fine-tuning paradigms cause the model to lose its pre-trained semantic knowledge. Consequently, their feature representations degrade into a narrow subspace suited only for specific binary tasks. In contrast, our results validate the effectiveness of the balanced objective, which successfully preserves rich semantic representations while learning forensic cues, thereby solving the forgetting problem.

In addition, we examine the t-SNE visualization of feature spaces from two perspectives: 1) comparing pre-trained feature spaces (Fig. 3(a), (c)) of CNNDet and PoundNet, and 2) analyzing their evolution (Fig. 3(b), (d)) after fine-tuning on

TABLE III
PERFORMANCE OF THE PROPOSED POUNDNET AND 10 STATE-OF-THE-ART METHODS ON 10 PUBLIC DEEPPAKE DETECTION BENCHMARKS IN TERMS OF AP, F1, ACC, AND AUC_{f1} . **BOLD** AND UNDERLINED HIGHLIGHT THE BEST AND THE SECOND-BEST PERFORMANCE, RESPECTIVELY.

Methods	ForenSynths				GenImage				GANGen-Detection			
	AP	F1	ACC	AUC_{f1}	AP	F1	ACC	AUC_{f1}	AP	F1	ACC	AUC_{f1}
CNNDet	89.05	54.04	70.18	53.59	64.95	9.58	52.04	10.38	82.69	37.27	62.15	37.57
FreDect	67.48	55.64	62.78	49.63	<u>78.49</u>	44.77	63.60	42.92	63.59	21.33	54.81	24.74
GramNet	57.52	14.30	50.10	14.57	63.01	33.06	<u>59.09</u>	33.10	50.00	0.41	49.99	0.77
Fusing	91.21	42.63	65.84	43.07	73.88	6.03	51.64	7.15	90.55	58.54	71.55	57.12
LNP	64.89	15.88	53.00	16.63	58.52	10.65	52.01	11.35	60.25	9.52	51.05	10.98
SPrompts	90.19	39.73	63.55	40.04	76.79	11.67	53.44	12.58	88.45	24.02	57.55	24.75
UnivFD	<u>92.96</u>	<u>60.49</u>	<u>74.69</u>	<u>59.73</u>	77.59	30.39	60.22	31.42	<u>91.91</u>	<u>79.95</u>	<u>83.68</u>	<u>75.87</u>
LGrad	53.12	3.90	<u>48.57</u>	4.46	51.93	2.12	50.30	2.63	<u>52.19</u>	0	49.99	0.37
NPR	47.60	12.98	50.95	13.19	55.28	21.79	53.69	21.92	48.07	47.65	49.97	47.73
Freqnet	53.12	8.58	50.38	9.00	53.83	3.55	50.20	4.02	49.62	49.49	51.04	49.37
PoundNet	94.09	79.21	80.37	77.65	80.98	47.77	68.02	47.68	93.22	85.79	84.49	83.67

Methods	DiffusionForensics				Ojha				DIF			
	AP	F1	ACC	AUC_{f1}	AP	F1	ACC	AUC_{f1}	AP	F1	ACC	AUC_{f1}
CNNDet	64.70	11.14	57.31	11.60	63.41	6.80	51.31	7.50	74.31	34.65	62.50	34.61
FreDect	45.53	<u>34.48</u>	49.16	<u>31.71</u>	68.14	54.99	62.04	50.49	76.76	55.08	67.57	50.98
GramNet	77.71	28.89	60.31	28.99	52.89	2.32	50.00	2.65	49.87	9.73	43.45	10.00
Fusing	62.89	1.57	56.23	2.77	73.61	7.71	51.88	9.53	81.64	33.92	63.36	34.01
LNP	79.50	22.58	61.04	22.86	46.45	7.84	49.80	8.39	56.42	10.46	50.07	11.23
SPrompts	81.01	12.97	57.38	13.41	71.93	12.79	53.76	13.53	79.02	29.71	59.97	30.07
UnivFD	62.42	18.39	60.68	19.65	93.87	<u>67.45</u>	<u>76.19</u>	<u>65.54</u>	88.31	<u>56.39</u>	<u>73.72</u>	<u>55.92</u>
LGrad	67.90	4.88	56.81	5.33	38.07	0.20	<u>49.86</u>	<u>0.62</u>	50.36	<u>0.47</u>	49.92	0.97
NPR	56.78	10.12	53.26	10.31	44.57	4.95	50.42	5.28	45.75	8.68	47.68	8.90
Freqnet	51.63	4.93	56.27	5.26	50.82	2.71	50.01	3.22	51.71	3.39	50.22	3.91
PoundNet	63.48	42.91	67.24	42.41	<u>91.98</u>	81.79	82.90	79.46	88.76	70.32	79.23	69.17

Methods	Celeb-DF-v1/v2				UADFV				AntiFake
	AP	F1	ACC	AUC_{f1}	AP	F1	ACC	AUC_{f1}	ACC
CNNDet	64.40/87.60	0.14/0.19	33.94/13.57	0.72/0.92	61.65	0.13	50.42	0.65	16.63
FreDect	72.54/88.10	<u>53.93/55.18</u>	51.88/43.61	<u>46.57/51.06</u>	62.42	<u>52.11</u>	60.03	<u>44.37</u>	27.46
GramNet	67.44/88.43	0.04/0.08	33.94/13.53	0.44/0.55	39.31	0	49.77	0.33	30.40
Fusing	61.87/86.66	0.21/0.56	33.86/13.70	1.32/2.23	53.18	0.65	50.49	1.69	15.72
LNP	64.16/86.76	1.66/3.30	34.11/14.75	2.17/3.96	44.87	10.18	51.73	10.34	20.06
SPrompts	66.38/87.30	1.75/1.94	34.33/14.29	2.53/2.82	46.20	11.72	48.01	12.34	22.84
UnivFD	83.73/92.49	16.64/10.28	39.66/18.10	20.54/14.96	91.50	43.08	63.87	44.23	38.19
LGrad	65.98/87.33	0.14/0.56	33.95/13.71	0.60/1.16	44.47	0.13	49.90	0.87	15.50
NPR	62.13/86.15	7.28/7.48	34.45/16.23	7.59/7.88	43.40	10.78	50.42	11.11	26.64
Freqnet	64.60/86.11	18.85/17.32	36.67/20.41	19.13/17.77	44.55	11.44	45.05	11.89	16.49
PoundNet	<u>83.69/93.16</u>	71.28/76.95	65.93/65.69	65.69/71.35	<u>85.92</u>	79.17	77.86	71.41	48.24

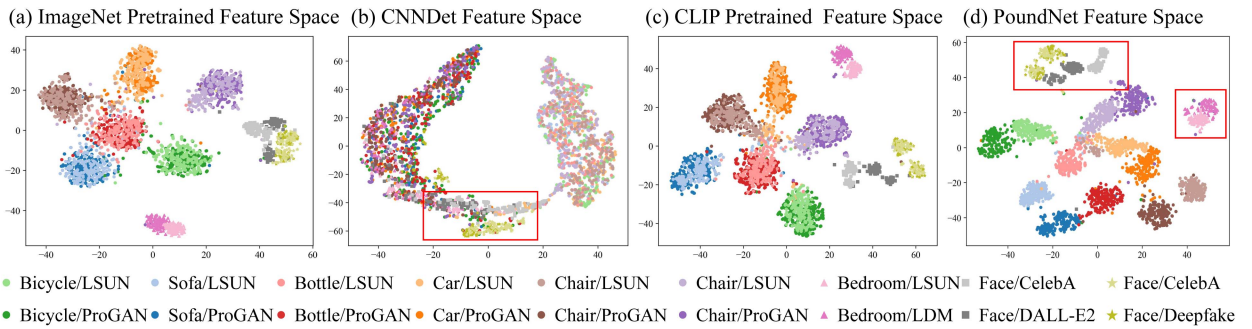


Fig. 3. Feature spaces for in-domain deepfakes (produced by ProGAN) and three unseen deepfakes (generated by LDM, DALL-E2, and Deepfake). Reals: [CLASS]/LSUN, [CLASS]/CelebA.

the same deepfake detection dataset. We use in-domain test data from ProGAN [9] and out-of-domain unseen deepfakes (generated by LDM [1], DALL-E 2 [3] and Deepfake [73]) to assess generalization. As shown in Fig. 3, CNNDet learns distinct feature representations for in-domain data but struggles

with unseen deepfakes, indicated by overlapping clusters of real and fake samples. This suggests CNNDet's features are overly specialized to the training data. In contrast, PoundNet preserves the structure of the pre-trained feature space while adapting to deepfake detection, resulting in better generalization to unseen

TABLE IV
IMPACT OF THE TWO MAIN HYPERPARAMETERS λ_1 AND λ_2 DEEFAKE DETECTION ON THREE USED DATASETS

λ_1	λ_2	ForenSynths				DIF				Ojha				Averge			
		AP	F1	ACC	AUC _{F1}	AP	F1	ACC	AUC _{F1}	AP	F1	ACC	AUC _{F1}	AP	F1	ACC	AUC _{F1}
0	0	90.13	69.89	78.89	62.31	85.35	52.18	70.96	50.48	90.47	53.16	63.61	51.63	88.65	58.41	71.15	54.81
0	1	95.30	74.42	81.25	72.69	87.61	58.45	74.51	57.54	93.18	61.16	72.84	59.70	92.03	64.68	76.20	63.31
1	0	93.95	65.72	75.92	64.14	85.90	54.35	71.65	53.07	89.31	56.34	69.80	54.17	89.72	58.80	72.46	57.13
1	1	94.09	79.21	80.37	77.65	88.76	70.32	79.23	69.17	91.98	81.79	82.90	79.46	91.61	77.11	80.83	75.43
0.1	0.1	94.93	78.87	85.11	77.14	90.03	63.12	76.95	62.36	92.12	69.93	77.26	67.90	92.36	70.64	79.77	69.13
0.1	1	95.18	75.23	82.76	73.34	88.53	60.28	75.56	59.54	90.68	62.28	72.94	60.74	91.46	65.93	77.09	64.54
1	0.1	94.20	77.84	81.90	75.56	86.53	62.03	75.72	60.55	93.63	70.02	77.79	67.49	91.45	69.96	78.47	67.87
0.5	0.5	94.76	74.34	81.08	72.87	88.58	60.48	75.21	59.74	91.68	63.62	73.89	61.73	91.67	66.15	76.73	64.78
0.5	1	94.35	79.29	82.93	77.56	87.41	65.56	77.28	64.74	92.85	75.29	79.96	72.89	91.54	73.38	80.06	71.73
1	0.5	94.42	68.92	77.03	67.85	84.21	59.87	73.75	58.74	91.10	62.05	72.76	60.19	89.91	63.61	74.51	62.26
2	2	93.56	76.04	82.19	74.14	85.72	60.91	74.34	60.04	86.23	61.43	71.75	58.95	88.50	66.13	76.09	64.38
2	1	92.89	69.80	77.38	67.83	83.56	57.70	72.17	56.17	89.51	57.53	70.61	55.58	88.65	61.68	73.39	59.86
1	2	91.33	73.07	81.55	72.01	82.83	61.32	76.32	60.42	90.34	72.35	78.70	70.13	88.17	68.91	78.86	67.52

TABLE V
INFLUENCE OF PROMPT DEPTH J AND PROMPT LENGTH M FOR DEEFAKE DETECTION ON THREE USED DATASETS

J	Test Set AP/ACC			
	ForenSynths	DIF	Ojha	Average
1	94.11/84.54	85.89/76.90	86.49/73.46	88.83/78.30
2	91.99/78.72	88.49/75.54	92.14/78.60	90.87/77.62
4	94.39/76.82	89.77/73.92	90.89/72.13	91.68/74.29
6	93.12/79.65	89.15/77.37	90.24/78.37	90.84/78.46
8	94.09/80.37	88.76/79.23	91.98/82.90	91.61/80.83

M	Test Set AP/ACC			
	ForenSynths	DIF	Ojha	Average
1	90.90/73.31	73.90/62.57	76.43/69.25	80.41/68.38
2	93.61/81.21	88.95/78.3	87.52/80.64	90.03/80.05
4	94.13/84.12	85.43/73.81	89.81/75.27	89.79/77.73
8	92.01/82.85	86.48/77.78	91.11/83.34	89.87/81.32
16	94.09/80.37	88.76/79.23	91.98/82.90	91.61/80.83

deepfakes. This is evidenced by more distinct clusters of real and fake samples in both in-domain and out-of-domain data, indicating PoundNet's ability to learn robust and transferable features.

E. Ablation Analysis

Impact of the Main Components: This ablation study aims to validate the effectiveness of the main components of the proposed PoundNet. We train PoundNet on ForenSynths [4] and test it on three datasets: ForenSynths, DIF, and Ojha, evaluating performance in terms of AP, F1, ACC, and AUC_{F1}. The most important components of PoundNet are the semantic-preserving term $\mathcal{L}_{spm}$ and class-aware binary term $\mathcal{L}_{cab}$. Table IV gives the study on the influence of the two hyperparameters λ_1 and λ_2 . When both parameters are set to 0, the model achieves moderate performance across all datasets. The performance notably improves when introducing non-zero values for both parameters. The optimal configuration appears when $\lambda_1 = 1$ and $\lambda_2 = 1$, yield the best average performance.

Impact of the Hyperparameters: Table V studies the effects of different prompt depths (J) and prompt lengths (M) on the model's performance. The study is measured across the ForenSynths, DIF, and Ojha test sets, and presented as Average Precision (AP) and Accuracy (ACC). While deeper prompts and longer prompts tend to yield slightly better results, the model's

TABLE VI
INFLUENCE OF THE NUMBER OF CLASSES USED FOR TRAINING

Class Num.	Test Set AP/ACC			
	ForenSynths	DIF	Ojha	Average
2	90.90/73.31	86.23/72.52	90.70/68.86	89.28/71.56
4	92.42/79.87	88.08/77.92	93.09/77.88	91.20/78.56
8	93.62/78.61	86.69/75.35	90.91/72.80	90.41/75.59
16	93.26/80.83	85.95/76.45	87.57/72.86	88.93/76.71
20	94.09/80.37	88.76/79.23	91.98/82.90	91.61/80.83

TABLE VII
PERFORMANCE OF DIFFERENT PRE-TRAINED MODELS. OA DENOTES OPENAI, WHILE OC REPRESENTS OPENCLIP.

Weights	Test Set AP/ACC			
	ForenSynths	DIF	Ojha	Average
OA ViT-B/16	86.04/74.33	85.44/78.12	87.75/73.82	86.41/75.42
OA ViT-L/14	94.09/80.37	88.76/79.23	91.98/82.90	91.61/80.83
OC ViT-L/14	93.46/79.67	87.63/77.32	91.73/82.09	90.94/79.69

performance does not drastically fluctuate with changes in different parameters, indicating robustness to these parameters.

Analysis of Training Class Number: Table VI shows the performance of PoundNet when trained with different numbers of classes from the ForenSynths training set, which contains a total of 20 classes. The results are presented as Average Precision (AP) and Accuracy (ACC) percentages across the ForenSynths, DIF, and Ojha test sets. Although PoundNet's performance generally improves with an increase in the number of classes, it remains relatively stable, indicating that the method is not sensitive to the number of classes used for training. This result reinforces the idea that the semantic classes act as general anchors to guide the learning of a robust representation, rather than requiring an exhaustive set of labels to function effectively.

Impact of Pre-trained Model Selection: Table VII compares the performance of various pre-trained models used to train PoundNet on the ForenSynths, DIF, and Ojha test sets. We keep all other hyperparameters the same, and only change the pre-trained weights of the CLIP architecture. We can see that larger models (ViT-L/14) generally offer better performance, with OpenAI ViT-L/14 being the best performer. This suggests that the increased capacity and feature representation power of larger models are beneficial for learning effective forensic detection features. However, PoundNet can get good performances with smaller models than other methods, demonstrating its efficiency

TABLE VIII
PERFORMANCE COMPARISON OF POUNDNET WITH DIFFERENT LOSS
FUNCTION COMBINATIONS

Objective	AP/F1/ACC/AUC _{f1}
Eq. 2 (Real/Fake)	85.68/55.55/70.49/52.33
Eq. 2 (Real/Fake [Class])	88.65/58.41/71.15/54.81
Eq. 2 + Eq. 4	89.72/58.80/72.46/57.13
Eq. 2 + Eq. 5	92.03/64.68/76.20/63.31
Eq. 2 + Eq. 4 + Eq. 5	91.61/77.11/80.83/75.43

TABLE IX
COMPARISON OF PERFORMANCES USING GROUNDTRUTH(GT), PREDICTED
LABELS(PD), NaN MEANS WE DON'T PROVIDE SEMANTIC CLASS
INFORMATION DURING INFERENCE

Train	Test	GenImage (AP/AUC/F1/ACC)	DiffusionForensics (AP/AUC/F1/ACC)
GT	NaN	80.98/82.12/47.77/68.02	66.04/73.46/50.39/72.26
GT	PD	76.82/78.37/57.93/68.36	77.10/83.16/59.94/75.03
PD	NaN	69.12/76.97/54.95/65.01	73.09/78.46/26.39/65.92
PD	PD	66.99/67.28/58.22/68.32	78.27/79.02/32.83/73.37

and adaptability in leveraging pre-trained knowledge, even with more compact models.

Impact of the Loss Function Combinations: Table VIII demonstrates how different configurations of the proposed objectives affect both AI-generated image detection and image classification performance. The balanced objective comprises three components: a class-agnostic binary loss for deepfake detection (2), a class-aware term to preserve pre-trained class semantics (4), and a class-aware binary term for within-class deepfake detection (5). The class-aware term (4) aims to balance the deepfake detection training against the valuable knowledge acquired from the training tasks learned from the pre-trained CLIP. And the class-aware term (5) to smoothly transfer the pre-trained CLIP from the general classification tasks to AI-generated image detection. Overall, our balanced objective decreases the risk of overfitting the training data to improve the AI-generated image detection generalization by balancing the pre-trained CLIP model's generalization capability.

Predicted vs. Groundtruth Labels: As there are a large amount of AI-generators based on known classes (like ProGAN-generated images [4] simulating LSUN 20 classes, or other deepfakes [12] generated by simulating ImageNet 1000 classes), using additional semantic class labels to train PoundNet is not costly. It is also feasible to use pseudo labels generated by some off-the-shelf deep learning models (like CLIP which could perform well for zero-shot image classification). To justify this, we conducted an experiment using labels predicted by CLIP instead of groundtruth labels to train PoundNet.

We employ ForenSynths as our training dataset and evaluate on two test sets: GenImage and DiffusionForensics. GenImage shares the same label space as ImageNet, from which its real data is sourced, while DiffusionForensics uses the LSUN bedroom subset as real data, ensuring reliable groundtruth labels for all samples. Table IX presents the model's performance under various training and testing configurations. The results reveal that although using predicted labels for both training and testing leads to a moderate performance decline, the model maintains reasonable effectiveness. The results in the table also demonstrate that training PoundNet does not require groundtruth labels

TABLE X
PERFORMANCE EVALUATION ACROSS MULTIPLE RANDOM SEEDS

Method	AP	AUC	F1	ACC
Ojha	92.82 ± 0.92	92.57 ± 0.91	77.04 ± 6.90	81.27 ± 3.33
ForenS.	95.32 ± 1.07	95.03 ± 0.84	81.33 ± 3.20	84.86 ± 4.19
DIF	90.24 ± 1.39	90.91 ± 1.42	69.24 ± 1.60	79.65 ± 1.24

to be effective. Crucially, the rows where the test protocol is NaN (signifying no class information was provided during inference) demonstrate strong performance, directly confirming the effectiveness of our class-agnostic inference strategy. Note that in our main experiments, we do not utilize class label information as additional input, as this would incur extra computational overhead during inference.

Robustness Analysis with Multiple Random Seeds: To rigorously validate PoundNet's performance stability, we conducted experiments across three random seeds, following recent best practices in deep learning evaluation. While many state-of-the-art deepfake detection methods [13], [14], [35], [37] typically report results from a single run, we performed three independent training runs with different random initializations to assess the statistical reliability of our approach.

Table X presents the comprehensive evaluation metrics (mean ± standard deviation) across these runs. The results demonstrate PoundNet's robust performance, with consistently small standard deviations across all metrics, indicating stable behavior regardless of initialization conditions.

V. CONCLUSION AND OUTLOOK

This paper identifies a "Penny-Wise and Pound-Foolish" phenomenon in AI-generated image detection and proposes PoundNet, which uses a learnable prompt design and a balanced objective to address this issue. Evaluation on the largest test sets to date (comprising 10 AI-generated image detection benchmark datasets and 5 object classification datasets) demonstrates that PoundNet achieves a remarkable 19% improvement in deepfake detection compared to state-of-the-art methods. Additionally, PoundNet maintains strong performance in object classification, achieving a 63% accuracy, while other AI image detection methods typically fail in this area.

For future work, we believe the proposed learning scheme can be applied to various other domains that involve transferring pre-trained deep learning models from upstream to downstream tasks. This approach would shed light on the direction for continuous learning across a wide range of task categories. Furthermore, the current learning scheme has been tested in two continual learning sessions (pre-training for zero-shot object classification and fine-tuning on a single AI-generated image dataset for deepfake detection). Future work will explore extended continual learning sessions.

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